

Module 1: Network + Network Fundamentals and building networks

Day 4 Firewall Basics

Task 1:

Blocking Spotify with an outbound rule

New Outbound Rule Wizard

Program

Specify the full program path and executable name of the program that this rule matches.

Steps:

- Rule Type
- Program
- Action
- Profile
- Name

Does this rule apply to all programs or a specific program?

☐ **All programs**
Rule applies to all connections on the computer that match other rule properties.

☒ **This program path:**
%USERPROFILE%\Downloads\AnyDesk.exe
Example: c:\path\program.exe
%ProgramFiles%\browser\browser.exe

Action

Specify the action to be taken when a connection matches the conditions specified in the rule.

Steps:

- Rule Type
- Program
- Action
- Profile
- Name

What action should be taken when a connection matches the specified conditions?

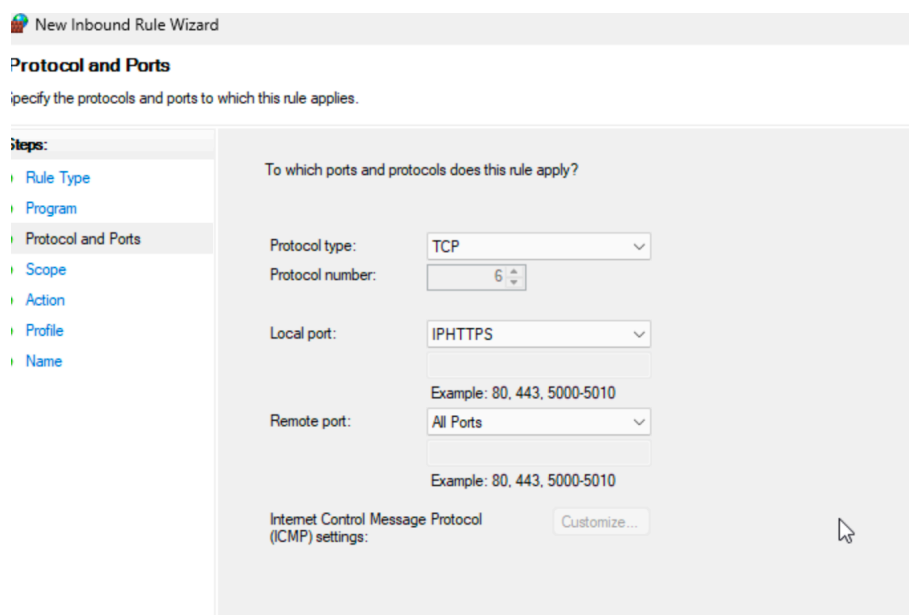
☐ **Allow the connection**
This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

☒ **Block the connection**

Task 2:

Allowing Chrome inbound only on port 443



The screenshot shows the 'New Inbound Rule Wizard' window in Windows Firewall. The 'Protocol and Ports' step is selected in the left-hand 'Steps' pane. The main area is titled 'To which ports and protocols does this rule apply?'. It contains the following settings:

- Protocol type: TCP (selected from a dropdown)
- Protocol number: 6 (displayed in a text box)
- Local port: IPHTTPS (selected from a dropdown)
- Example: 80, 443, 5000-5010 (displayed below the local port dropdown)
- Remote port: All Ports (selected from a dropdown)
- Example: 80, 443, 5000-5010 (displayed below the remote port dropdown)
- Internet Control Message Protocol (ICMP) settings: A section with a 'Customize...' button.

Task 3:

Blocking all traffic from a specific IP (123.45.67.89)

Steps:

1. Open Windows Defender Firewall with Advanced Security

Press Win + R, type wf.msc, and hit Enter.

2. Create a new rule (both directions if needed)

Since IP blocking usually needs to cover both directions, you'll create this rule twice — once under Inbound Rules and once under Outbound Rules — or create one rule and duplicate it. Click New Rule in either section to start.

3. Choose 'Custom' as the rule type

Select Custom, then click Next through the Program screen leaving it set to 'All programs', since you want to block this IP regardless of which app is talking to it.

4. Leave protocol and port settings as default

Leave Protocol type as 'Any' unless you specifically want to block only certain ports from that IP; click Next.

5.Go to the Scope screen and add the IP address

This is the key step. Under 'Which remote IP addresses does this rule apply to?', select 'These IP addresses', click Add, and enter 123.45.67.89. This restricts the rule to only that specific address.

6.Set the action to Block the connection

Choose 'Block the connection' and proceed through the profile screens, leaving all checked.

7.Name the rule and finish

Name it something like 'Block IP 123.45.67.89' and click Finish. Repeat the same process for the opposite direction (inbound or outbound, whichever you didn't just create) to fully block traffic to and from that address.

Task 4:

	Hardware Firewall	Software Firewall
Placement & scope	Sits between the entire network and the internet as a separate physical device, protecting every connected machine at once	Installed on an individual computer, protecting only that one device
Performance Impact	Doesn't use the host computer's CPU/RAM — dedicated hardware handles filtering, so no slowdown on individual machines	Consumes some of the host computer's own resources to inspect traffic
Configuration and flexibility	Centrally managed by IT staff — one config change applies network-wide, but less granular control per-app	Configured per-device, allowing fine-grained, per-application rules (like blocking just Spotify) but requiring separate setup on every machine